

Problem-Solution Fit

Date	27 June 2026
Team ID	
Project Name	Raising waters
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] Disaster Management Authorities	6. CUSTOMER LIMITATIONS Dependence on manual monitoring Slow communication of alerts	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Easily accessible Low implementation cost Delayed predictions
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Late flood warnings Loss of human lives and property Inaccurate flood prediction methods	9. PROBLEM ROOT / CAUSE [RC] Climate Change Heavy rainfall River overflow Poor drainage systems	7. BEHAVIOR + ITS INTENSITY [BE] Monitor rainfall during monsoon season Check weather updates frequently
3. TRIGGERS TO ACT [TR] Heavy rainfall Dense cloud cover High annual rainfall	10. YOUR SOLUTION [SL] Develop a machine learning-based flood prediction and early warning system using random forest, K-nearest neighbours(KNN). This system analysis historical weather data, annual rainfall to predict flood risk.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Flask Web Application IBM Cloud Deployment
4. EMOTIONS BEFORE Fear, Anxiety, Panic , Uncertainty AFTER Confidence , Safety, Preparedness		OFFLINE District Disaster Management offices Emergency Control Rooms Local Administration